



## INFORME EXPERT – INFORME EXPERTO EXTERNO – EXTERNAL REFEREE REPORT

Nom de l'expert / Nombre del experto / Name of the referee:

Categoria / Categoría / Category:

Departament, Universitat a què pertany / Departamento, Universidad a la que pertenece /  
Department, University to which s/he belongs:

Títol de la tesi oficial / Título de la tesis oficial / Title of the official thesis:

Hardware-Software Co-Design for Accelerating Sparse Tensor Algebra

Nom del doctorand que presenta la tesi:

MARCO SIRACUSA

**Especifiqueu les raons que avalen la qualitat de la tesi per a la seva defensa pública:**

Objectius assolits; Originalitat del treball; Metodologia emprada; Hipòtesis contrastades; Qualitat del manuscrit

**Especificar los motivos que avalan la calidad de la tesis mencionada para su defensa pública:**

Objetivos logrados; Originalidad del trabajo; Metodología usada; Hipótesis contrastadas; Calidad del manuscrito

**Specify reasons endorsing the quality of the above-mentioned thesis for its public reading:**

Objectives achieved; Originality of the work; Methodology used; Hypotheses tested, quality of the manuscript

This work proposes an interesting approach to efficiently execute sparse applications. It describes the prior work, does a thorough job describing its implement and evaluating its performance, and finally describes a software system which can compile applications to the new architecture.

**Valoració absoluta i/o ponderada de la tesi en relació amb altres treballs d'investigació:**

**Valoración absoluta y/o ponderada de la tesis presentada en comparación con otros trabajos de investigación:**

**Absolute and/or relative assessment of the thesis in comparison with other works of research:**

Aquesta tesi està entre les millors / Esta tesis está entre las mejores / This thesis is among the top:  
Molt Bona / Muy Buena / Very Good

**Considera que la tesi esmentada és apta per al tràmit de lectura i la defensa pública?**

**¿Considera que la tesis anteriormente mencionada es apta para su lectura y defensa pública?**

**In consideration of all the above, is the thesis judged to be suitable for public reading?**

☐ Sí / Yes

☐ Sí amb correccions majors / Sí con correcciones mayores / Yes with major corrections

☒ Sí amb correccions menors / Sí con correcciones menores / Yes with minor corrections

☐ No

**Observacions / Observaciones / Observations:**

Overall I thought the technical work was of high quality and is a good PhD dissertation. Most of the comments in the attached document were places where I thought the writing of the document could be improved. I know I that have specific rules for things that I write, which might not be your rules, so you don't need to follow all my style rules. But in case you are interested, here are some guidelines which might be helpful to think about when you edit your thesis:

- Chapter/section intros that state what the following sections are, aren't that helpful. The point of the intro is to motivate the sections that follow. Done well, it informs the reader why the material in the thesis/chapter/section is there in the first place. Its goal is to provide the reader a thread which they can follow to help them understand the material

- o A reader has a much easier time reading the material when they know what to expect / where they are heading. Technical writing isn't supposed to be like reading a mystery where clues are given and the point is revealed later. In some of the technical sections, the details come first, and then the results are given later. When reading the details, it is not clear what problem is being addressed. Motivational inductions would help, since they should talk about what problem is being addressed, and what part of the problem is addressed in each section

- A reader should be able to understand the work and not read any of the equations, and consistent symbology is critical for a reader to follow the equations. In your thesis, I have marked a few places where I got confused by your equations because I didn't know what a variable or subscript represented. Being consistent by using the same value for the same parameter would help me. For example, always using 'l' to represent lane #.

- Using a consistent set of measurements is helpful for comparing results. I know this is not always possible, or useful, so you should explain why they change when they change. Also if you say that a point is critical, like # of outstanding memory requests, then I expect that number would be reported in the thesis.

- Conclusions are what you concluded (learned) from the work. It isn't a summary of the work, which is what you did. Today, unfortunately, it is the approach that most people take.